

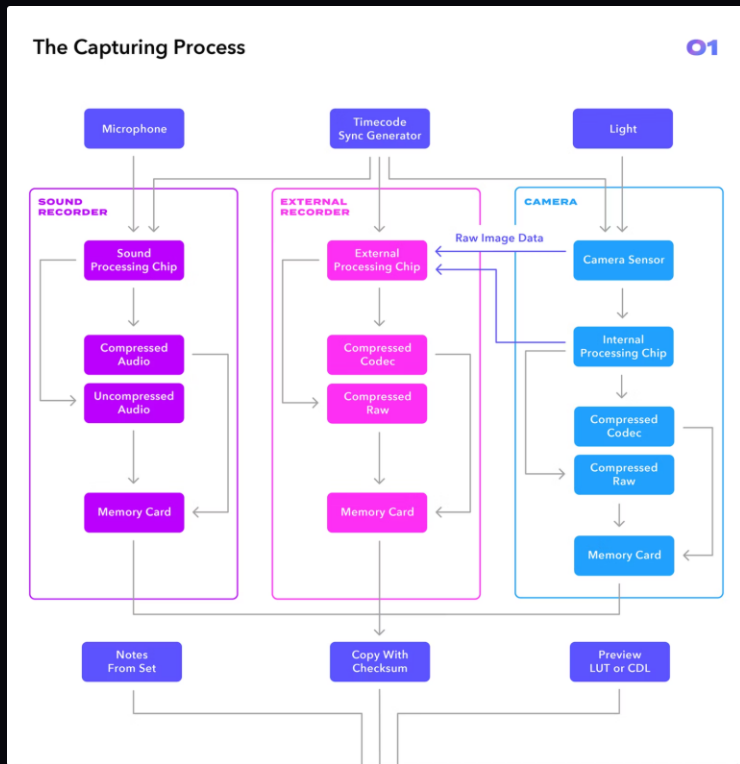
Quantization Noise in Digital Imaging

Identifying its Impact and Graphical Analysis

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Introduction to Quantization



Definition: Quantization maps continuous intensity values to discrete digital levels.

Role: Final step after sampling to produce pixel values for storage and processing.

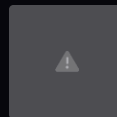
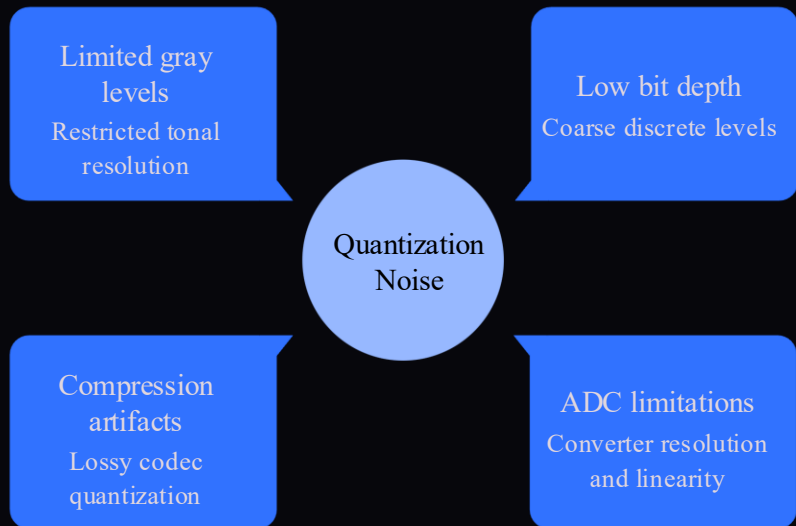
Sampling vs Quantization: Sampling fixes spatial coordinates; quantization discretizes amplitude (intensity).

Importance for CV: Affects feature extraction, visual fidelity, and algorithmic robustness.

Bit depth → Gray levels (examples)

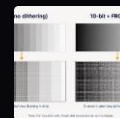
Bit depth	Gray levels
2-bit	4
4-bit	16
8-bit	256
16-bit	65536

Quantization Noise — What and Why



Causes

- Low bit depth (coarse levels)
- Analog-to-digital converter resolution and linearity
- Compression quantization (lossy codecs)
- Limited gray-level representation



Visual Example

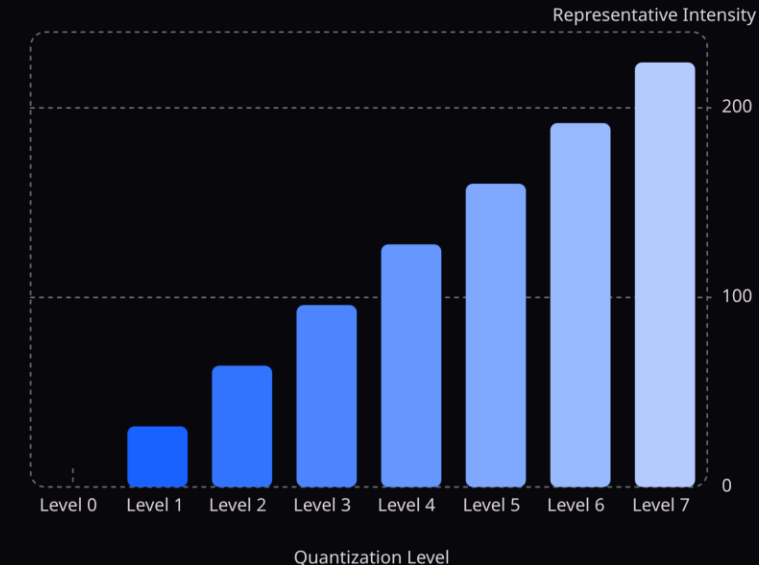
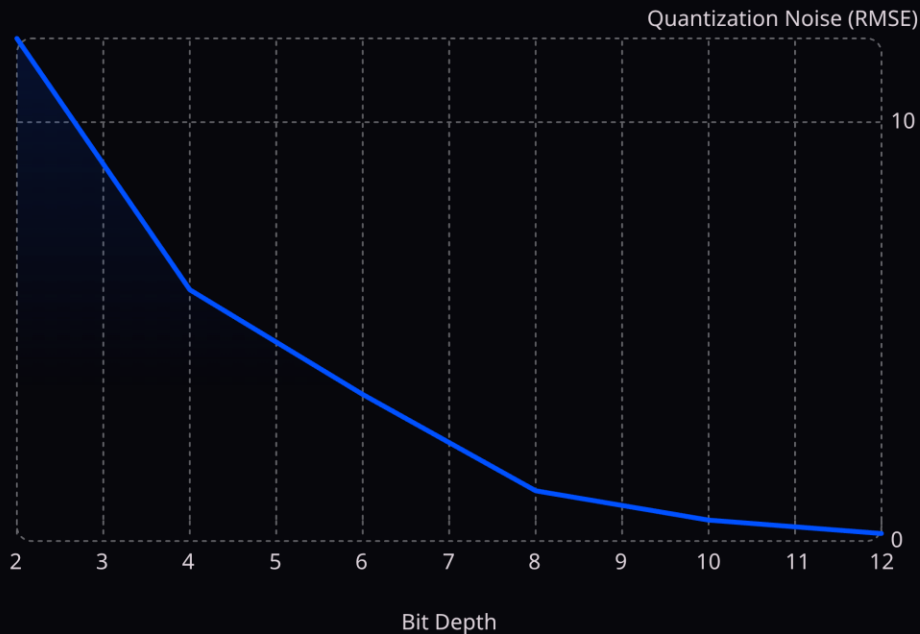
4-bit shows banding and stepped intensities;
8-bit preserves smooth gradients.

Definition (simple): Quantization Noise = Original Pixel Value – Quantized Pixel Value

Impact and Graphical Analysis

Effects on image quality

- Loss of fine details and texture
- Banding / contouring in smooth gradients
- Reduced contrast and dynamic nuance
- Poor feature extraction → degraded recognition accuracy



Staircase/quantization-level graph: visualizes discrete steps that produce banding in gradients.

Line chart (example): measured RMSE of quantization noise vs bit depth — noise falls rapidly with increased bits.

Conclusion: Higher bit depth significantly reduces quantization noise and visible banding.

Applications & Noise Reduction Techniques

Relevant Applications (icons below)



Medical Imaging

Diagnostic fidelity requires low-noise quantization.



Face Recognition

Feature details must be preserved for accurate matching.



OCR

Text strokes need high dynamic resolution to avoid misreads.



Autonomous Vehicles

Sensor quantization impacts perception and safety.



Satellite Imaging

Remote sensing requires fine radiometric precision.



Security Surveillance

Low-light quantization affects identification tasks.

Noise reduction strategies

1. Increase bit depth and use higher-resolution ADCs
2. Apply dithering to decorrelate quantization error from signal
3. Use image enhancement and adaptive contrast techniques
4. Design compression-aware quantizers (rate-distortion optimized)
5. Post-filtering (edge-preserving denoisers, bilateral filters)

Conclusion & Key Takeaways

1

Quantization converts continuous intensities into discrete levels

Essential for digital representation but introduces error.

2

Quantization noise is unavoidable but manageable

Design choices (bit depth, ADC, dithering) control its impact.

Higher bit depth → markedly better image fidelity

☐ Reduces RMSE and visible banding; improves CV performance.

References

- Gonzalez, R. C., & Woods, R. E., *Digital Image Processing* (4th Ed.)
- Szeliski, R., *Computer Vision: Algorithms and Applications* (2nd Ed.)